

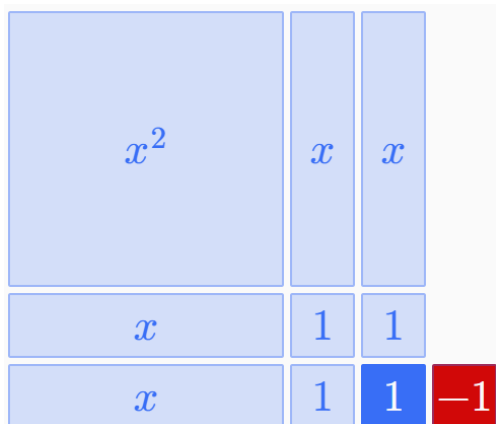
MATH-18A *Completing the Square*

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Visual Approach

Fill in the blanks of the square by adding “ h ”, and subtract to account for the stuff you added. Make sure you have net zero change!

$$x^2 + 4x + 3 = x^2 + 4x + 3 + 1 - 1 = (x + 2)^2 - 1$$



Workbook Ex. 2a

$$f(x) = x^2 - 2x - 3$$

$$h = -b/2a = -(-2)/(2 \cdot 1) = -1$$

add h^2 and $-h^2$:

$$f(x) = x^2 - 2x + (-1)^2 - (-1)^2 - 3$$

rewrite as squared form:

$$f(x) = (x - 1)^2 - (-1)^2 - 3$$

$$\text{simplify: } f(x) = (x - 1)^2 - 2$$

Workbook Ex. 2b

$$f(x) = -x^2 - 4x + 1$$

factor out negative (to work with inside):

$$f(x) = -(x^2 + 4x - 1)$$

$$h = -b/2a = -(4)/(2 \cdot 1) = -2$$

add h^2 and $-h^2$:

$$f(x) = -(x^2 + 4x + (-2)^2 - (-2)^2 - 1)$$

rewrite as squared form:

$$f(x) = -((x + 2)^2 - (-2)^2 - 1)$$

$$\text{simplify: } f(x) = -(x + 2)^2 + 5$$

Formulaic Approach #1

$$a(x + b/2a)^2 + c - (b^2/4a)$$

same a, b, c as ABC form

Formula Approach #2

$$a(x - h)^2 + k$$

- $h = -b/2a$
- $k = f(h)$ plug in $x = h$ into equation

Workbook Ex. 4

$$f(x) = -3x^2 - 606x - 31000$$

$$h = -b/2a = -(-606)/(2(-3)) = -101$$

$$k = f(-101)$$

$$= -3(-101)^2 - 606(-101) - 31000$$

$$= -397$$

vertex is at $(-101, -397)$

$$f(x) = -3(x + 101)^2 - 397$$

Workbook Ex. 5

$$f(x) = 4x^2 - 600x + 22510$$

$$h = -b/2a = -(-600)/(2(4)) = 75$$

$$k = f(75)$$

$$= 4(75)^2 - 600(75) + 22510$$

$$= 10$$

vertex is at $(75, 10)$

$$f(x) = 4(x - 75)^2 + 10$$